Auto Grade Using Natural Language Processing

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ABSTRACT

- Built a system to **generate and evaluate subjective & objective answers** with NLP.
- **Subjective:** takes user responses, validates, and scores using NLP.
- **Objective:** evaluates predefined tasks automatically.
- **Results stored in Excel** for easy analysis.
- Improves **speed, accuracy, and efficiency** in handling large text data.

INTRODUCTION

- Rising need for systems that can **generate questions** and **evaluate subjective answers** with accuracy.
- Project provides a **complete solution** using advanced **NLP techniques**.
- Simplifies and improves **assessment of written responses**.
- Combines **powerful text-processing tools** with a **user-friendly interface**.
- Designed as a **practical tool for researchers and analysts**.
- Supports **both subjective and objective evaluations**.
- Workflow:
- Import required **libraries and datasets**.
- **Clean text data** for better processing.
- **Generate relevant questions** using **NLTK**.

OBJECTIVES

- **Generates & evaluates** subjective and objective answers efficiently.
- Uses **advanced NLP techniques** for accurate, streamlined assessment.
- Offers a **user-friendly interface** for researchers and analysts.
- Supports **automated text cleaning** and dual evaluation modes.
- Stores results systematically in **Excel** for insights & improvements.
- Efficiently **generates and evaluates** subjective & objective answers using NLP.
- Supports **automated text cleaning** and dual assessment modes.
- **Stores results** for easy analysis and actionable insights.

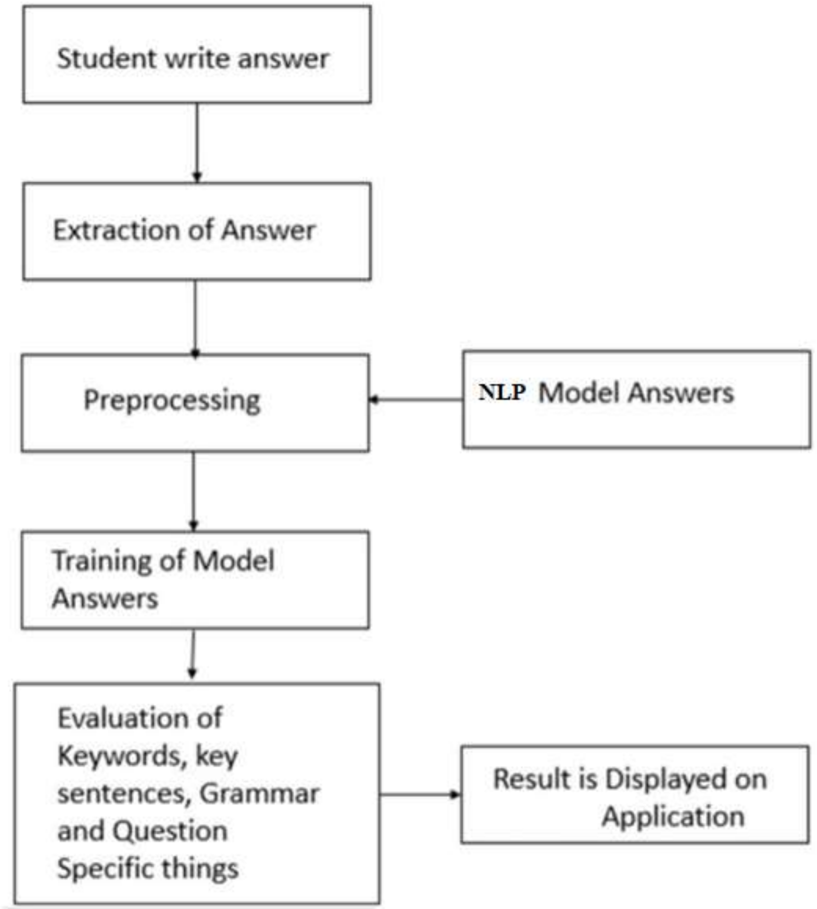
EXISTING SYSTEM

- Growing need for **efficient evaluation of textual data** highlights challenges in NLP.
- Existing methods often **fail to handle both subjective and objective assessments**.
- **Manual evaluation** is inefficient, less reliable, and reduces accuracy.
- Project develops a **comprehensive NLP-based system** to address these gaps.
- Combines **advanced text-processing algorithms** with a **user-friendly interface** to improve workflow, accuracy, and data analysis.

PROPOSED SYSTEM

- **Automates validation** and improves accuracy for fast, reliable insights.
- Evaluates both **subjective (user responses)** and **objective (automatic)** data using NLP.
- **Workflow:** import libraries & datasets, clean text, generate questions via **NLTK**.
- Applies **NLP algorithms** for accurate answer validation.
- **Stores results in Excel** for easy access and analysis.

SYSTEM ARCHITECTURE



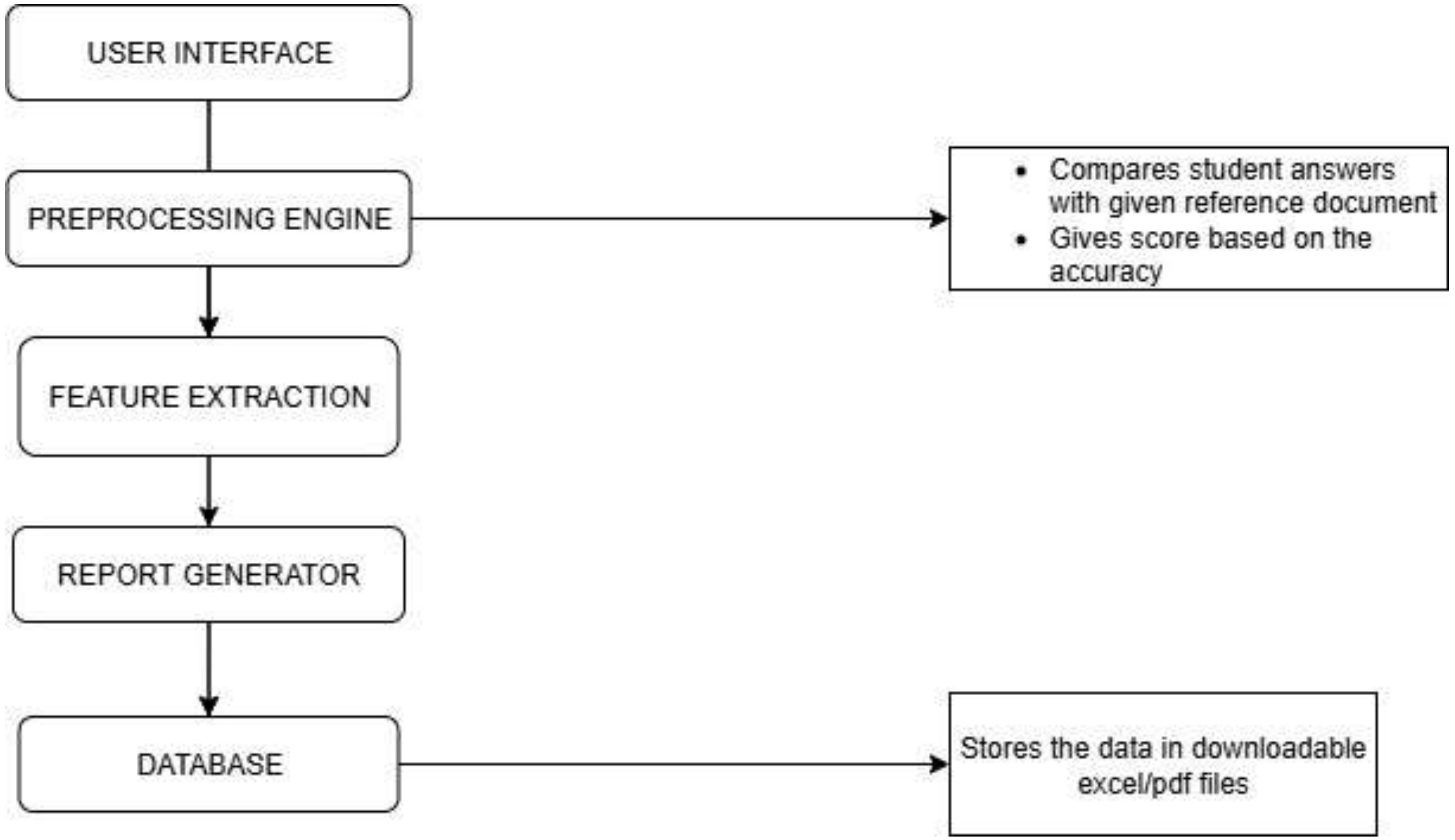
SOFTWARE REQUIREMENTS

- 1) Software : Anaconda
- 2) Primary Language : Python
- 3) Frontend Framework : Flask
- 4) Back-end Framework : Jupyter Notebook
- 5) Database : Sqlite3
- 6) Front-End Technologies : HTML, CSS, JavaScript and Bootstrap4

PHASE 1–REQUIREMENT GATHERING & ANALYSIS

- Identify Stakeholder End Users, Admins, Developers
- Define Supported File Types & Max Upload Size
- Set Performance Benchmarks (≤ 2 sec for short, ≤ 5 sec for long answers)
- Finalize Tech Stack
- Frontend: HTML, CSS, JavaScript (Bootstrap)
- Backend: Python
- Database: MySQL/PostgreSQL (production)
- Storage: Local in development, AWS S3 in production

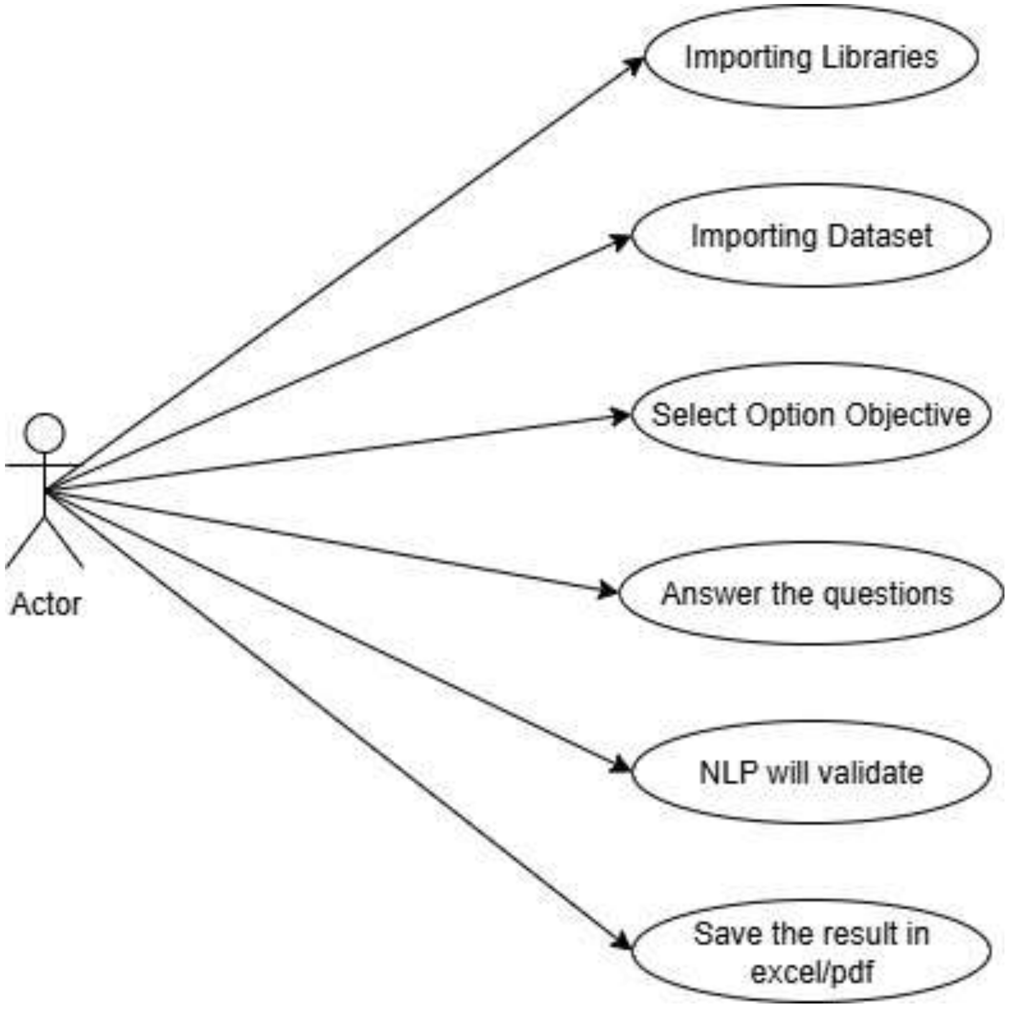
PHASE 2– System design – logical design



PHASE 3– TESTING

- **Unit Testing:** Preprocessing, TF-IDF, BERT/SBERT, Similarity modules.
- **Integration Testing:** End-to-end grading flow (submission → scoring → storage).
- **Performance Testing:** 500+ concurrent submissions, ≤3 sec grading time.
- **Accuracy Monitoring:** Track grading accuracy, false positives/negatives, target ≥85% match with human scoring.

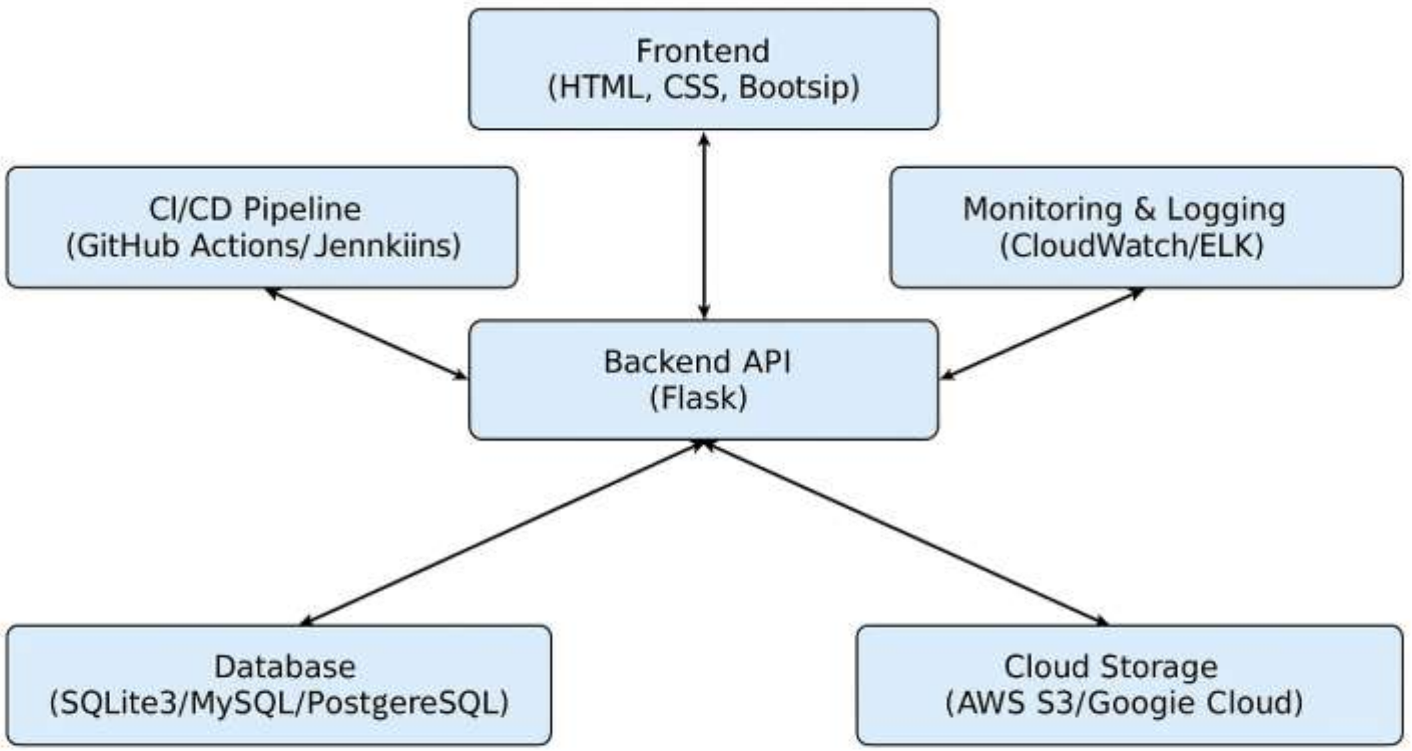
PHASE 2– System design – UML DIAGRAM



PHASE 4 – DEVELOPMENT

- Answer Submission Initiation
- Metadata Extraction & Preliminary Check
- Text Preprocessing
- Feature Extraction
- Similarity Analysis
- Scoring Logic.
- Feedback Generation
- Result Storage & Metadata Update

PHASE 5 – DEPLOYMENT



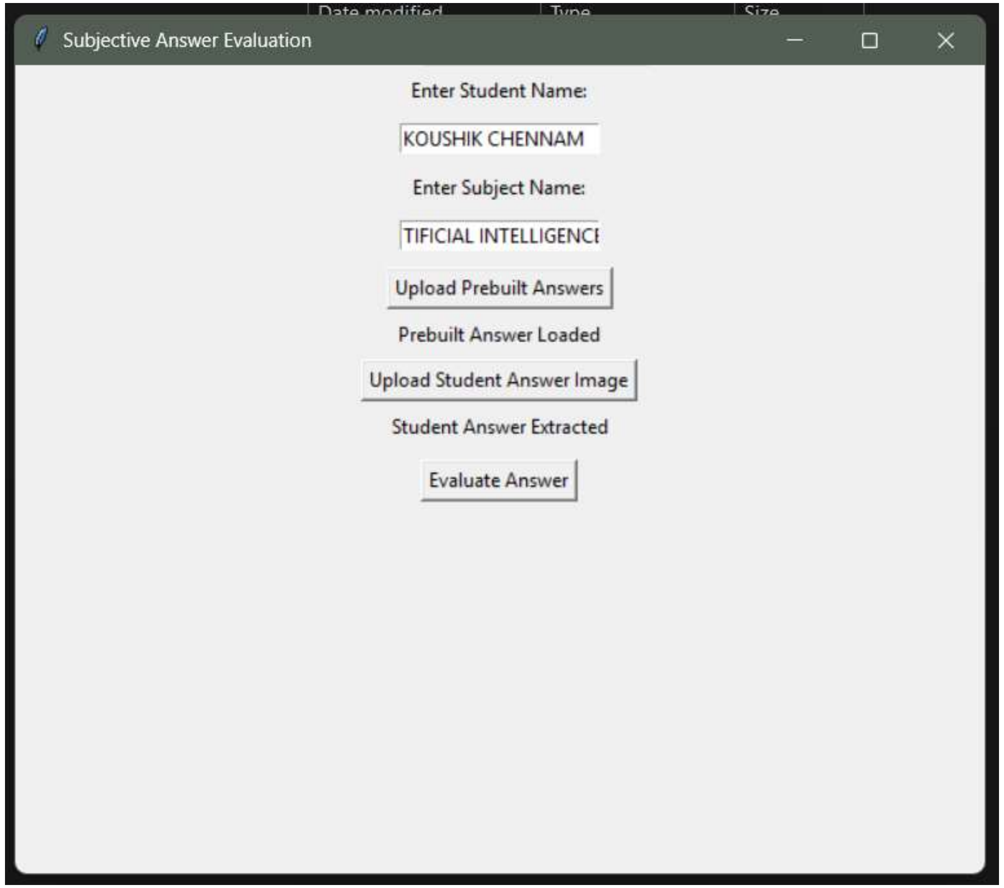
PHASE 6– MAINTENANCE AND FUTURE ENHANCEMENTS

- Monitor system performance & collect user feedback.
- Add support for more file types (e.g., audio, 3D models).
- Integrate blockchain for secure grading logs.
- Upgrade AI models (BERT → GPT-based) for higher accuracy.

IMPLEMENTATION

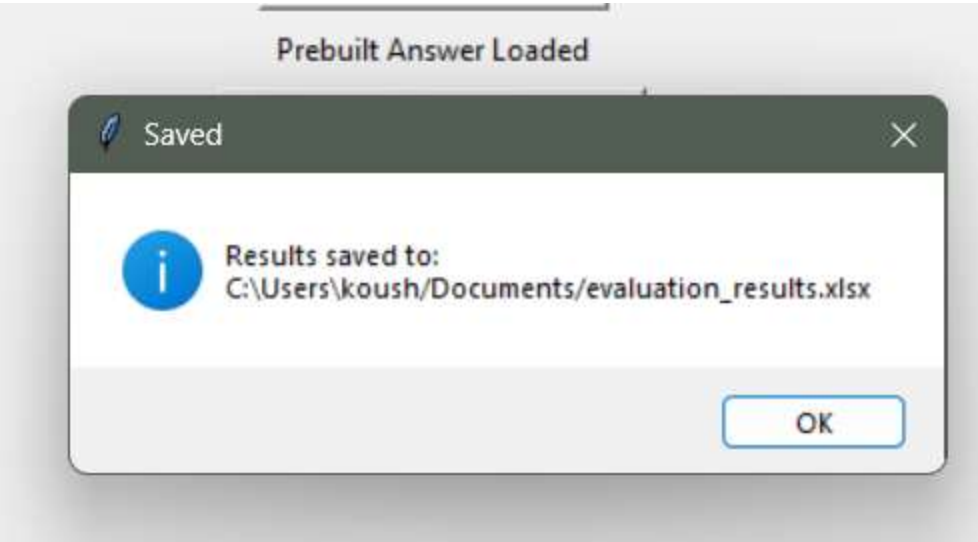
APPLICATION INTERFACE

- Step 1 : Enter the name
- Step 2 : Enter the subject name
- Step 3 : Upload the Prebuilt answers
- Step 4 : Upload Student answer image
- Step 5 : Evaluate the answer



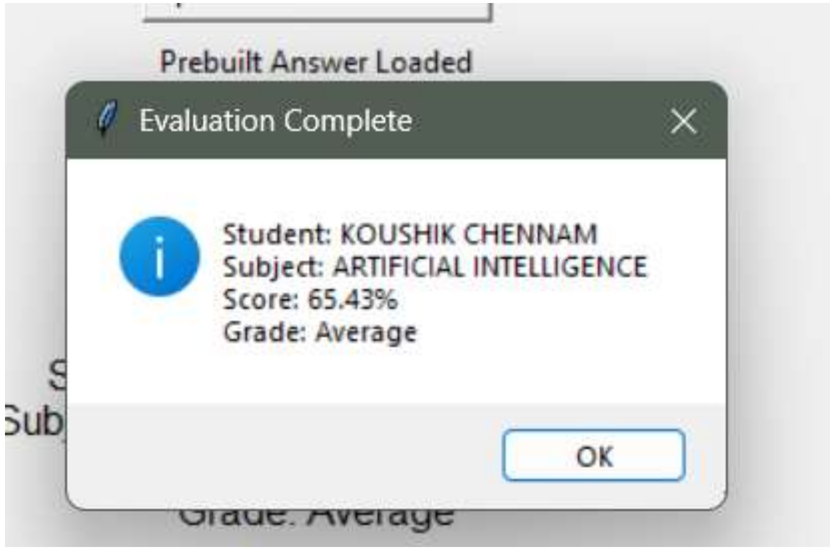
RESULTS

- After entering the details all the results will be saved to an excel file



RESULTS

- It displays the percentage and displays which grade student has obtained



RESULTS

- All the results are stored in an excel file to save the data

	A	B	C	D	E	F
1	**Student Name**	**Subject Name**	**Prebuilt Answer**	**Student Answer**	**Similarity Score (%)**	**Grade**
2	KOUSHIK	CSE	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
3	KOUSHIK	CHENNAM	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
4	KOUSHIK	CHENNAM	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
5	ARUN KUMAR	AI	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
6	KOUSHIK CHENNAM	C	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
7	K	C	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
8	KOUSHIK CHENNAM	ARTIFICIAL INTELLIGENCE	Q1. What is Artificial Intelligence? Prebuilt Answer: Artificial Intell	Q7. What is Artificial Intelligence?Student Ansv	65.43082484	Average
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CONCLUSION

- Develops a **robust and efficient evaluation system** in NLP.
- Combines **advanced NLP techniques** with a **user-friendly interface** for researchers and analysts.
- Implements **systematic workflow** for both subjective and objective assessments.
- Ensures **accuracy and reliability** using sophisticated NLP algorithms.
- **Stores results systematically** for easy access, analysis, and scalability.

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Thank You